Table 43. Flash memory characteristics (continued)

Symbol	Parameter[1]	Conditions	Min	Typ	Max	Unit
t_{prog_page}	Page (2 Kbyte) programming time	Normal programming	-	21.8	36.6	ms
		Fast programming	-	13.7	22.4	
t_{ERASE}	Page (2 Kbyte) erase time	-	-	22.0	40.0	ms
t_{prog_bank}	Bank (64 Kbyte[2]) programming time	Normal programming	-	0.7	1.2	s
		Fast programming	-	0.5	0.7	
t_{ME}	Mass erase time	-	-	22.1	40.1	ms
$I_{DD(FlashA)}$	Average consumption from V_{DD}	Programming	-	3.0	-	mA
		Page erase	-	3.0	-	
		Mass erase	-	5.0	-	
$I_{DD(FlashP)}$	Maximum current (peak)	Programming, 2 µs peak duration	-	7.0	-	mA
		Erase, 41 µs peak duration	-	7.0	-	

1. Specified by design. Not tested in production.
2. Values provided also apply to devices with less flash memory than one 64 Kbyte bank

Table 44. Flash memory endurance and data retention

Symbol	Parameter[1]	Conditions	Min	Unit
N_{END}	Endurance	T_J = -40 to +130 °C	10	kcycles
t_{RET}	Data retention	1 kcycle[2] at T_A = 85 °C	30	Years
		1 kcycle[2] at T_A = 105 °C	15	
		1 kcycle[2] at T_A = 125 °C	7	
		10 kcycles[2] at T_A = 55 °C	30	
		10 kcycles[2] at T_A = 85 °C	15	
		10 kcycles[2] at T_A = 105 °C	10	

1. Evaluated by characterization. Not tested in production..
2. Cycling performed over the whole temperature range.